

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5115

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 2

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Nalluri Charan Sai (192372003) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Nalluri Charan Sai

Annexure A

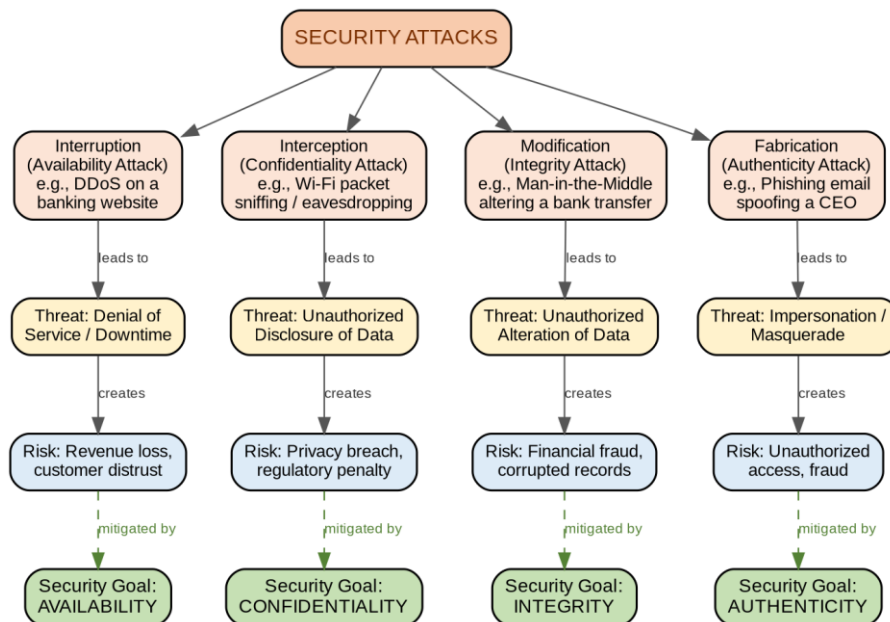
CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:

- Show how each attack leads to specific threats and risks**
- Map how security goals help mitigate them Provide one real-world example for each link**

This task connects four categories of security concepts along a single causal chain: a security attack is the actual hostile action carried out against a system; it exposes or realizes a threat, which is the potential danger the attack represents; if unaddressed, that threat translates into a concrete risk — a measurable business or operational consequence; and finally, a security goal (confidentiality, integrity, availability, and, by extension, authenticity) is the protective objective whose enforcement mitigates that specific risk. Understanding this chain is essential for CO1 because it shows why classical encryption techniques exist in the first place: every cipher, key, and protocol is ultimately built to defend one of these goals against a class of attacks.

(a) Concept Map Diagram



The diagram above traces four parallel chains, one per major attack category defined in the classical security-attack taxonomy (interruption, interception, modification, and fabrication), showing how each flows downward from Attack → Threat → Risk → mitigating Security Goal.

(b) How Each Attack Leads to a Specific Threat and Risk

1. Interruption → Denial-of-Service Threat → Availability Risk: An interruption attack destroys or disables a system resource so that legitimate users cannot access it when needed — for example, a Distributed Denial-of-Service (DDoS) attack that floods a banking website's

servers with traffic. The immediate threat is loss of service (downtime), and the resulting risk is direct revenue loss during the outage, damaged customer trust, and potential breach of Service-Level-Agreement (SLA) obligations. This chain is mitigated by the Availability goal, enforced through techniques such as redundancy, load balancing, and traffic-filtering firewalls.

2. Interception → Disclosure Threat → Confidentiality Risk: An interception attack allows an unauthorised party to gain access to data or a system without permission — for example, an attacker sniffing unencrypted packets on a public Wi-Fi network to capture login credentials. The threat here is unauthorised disclosure of sensitive information, and the resulting risk is privacy violation, identity theft, or regulatory penalties (such as under data-protection law) if personal data is exposed. This is mitigated by the Confidentiality goal, enforced through encryption of data in transit (e.g., TLS/HTTPS) and access controls.

3. Modification → Alteration Threat → Integrity Risk: A modification attack involves an unauthorised party not only accessing an asset but actually tampering with it — for example, a Man-in-the-Middle (MitM) attacker intercepting a bank transfer request and silently changing the destination account number. The threat is unauthorised alteration of data, and the risk is direct financial fraud, corrupted records, or cascading errors in downstream systems that trust the altered data. This is mitigated by the Integrity goal, enforced through mechanisms such as hash functions, digital signatures, and message authentication codes (MACs) that detect any tampering.

4. Fabrication → Impersonation Threat → Authenticity Risk: A fabrication attack involves an unauthorised party inserting counterfeit objects into a system — for example, a phishing email spoofed to appear as though it comes from a company's CEO, instructing an employee to wire funds. The threat is impersonation or masquerade, and the risk is fraudulent transactions, unauthorised system access, or reputational damage once the fabrication is discovered. This is mitigated by the Authenticity goal (closely tied to Integrity), enforced through digital certificates, digital signatures, and multi-factor authentication.

(c) How Security Goals Mitigate Each Risk

Each of the three (or, in extended models, four) core security goals maps directly onto one class of risk identified above, forming the protective half of the concept map:

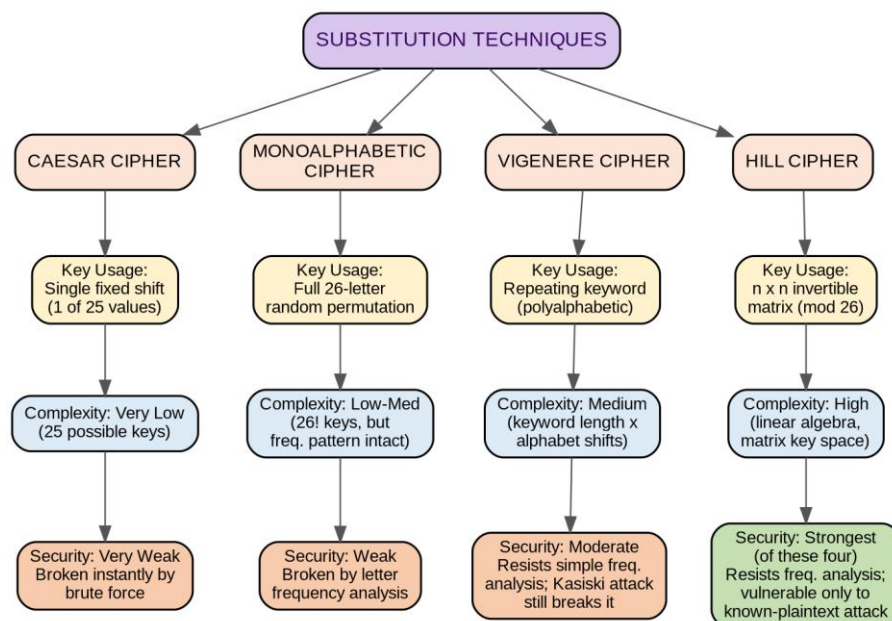
- Confidentiality ensures that information is disclosed only to authorised parties, directly countering interception-driven disclosure risk — implemented through encryption (symmetric ciphers such as AES, asymmetric ciphers such as RSA) and strict access-control policies.
- Integrity ensures that data cannot be altered undetected, directly countering modification-driven risk — implemented through cryptographic hash functions (SHA-256), digital signatures, and checksums.
- Availability ensures that systems and data remain accessible to legitimate users when needed, directly countering interruption-driven downtime risk — implemented through redundancy, failover clustering, rate limiting, and DDoS mitigation services.
- Authenticity (often treated as an extension of integrity) ensures that the claimed identity of a user, system, or message sender is genuine, directly countering fabrication-driven impersonation risk — implemented through digital certificates, public-key infrastructure (PKI), and multi-factor authentication.

Taken together, the concept map demonstrates that classical and modern cryptographic techniques are not applied in isolation; each technique is selected specifically because it enforces the security goal that neutralizes a particular attack-threat-risk chain, which is the core rationale behind encryption techniques for the ensuring of data security.

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

This task compares four classical substitution ciphers — Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher, and Hill Cipher — along three dimensions: how each uses its key, how computationally/mathematically complex each is, and how strong each is against cryptanalysis. Although all four fall under the umbrella of 'substitution' (each replaces plaintext letters with other letters or values rather than rearranging them, as transposition ciphers do), they differ enormously in sophistication, and this progression illustrates the historical evolution of classical cryptography from trivially breakable schemes toward mathematically grounded ones.

(a) Concept Map Diagram



(b) Comparison Summary Table

Technique	Key Usage	Complexity	Security Strength	Primary Weakness
Caesar Cipher	Single fixed shift value (1–25)	Very Low	Very Weak	Only 25 keys — brute force in seconds
Monoalphabetic Cipher	Full random 26-letter substitution table	Low–Medium	Weak	Preserves letter frequency — broken by frequency analysis
Vigenère Cipher	Repeating keyword (polyalphabetic shifts)	Medium	Moderate	Repeating key period found via Kasiski/Friedman tests
Hill Cipher	$n \times n$ invertible matrix (mod 26)	High	Strongest of the four	Linear — broken by known-plaintext attack (matrix algebra)

(c) Detailed Comparison by Dimension

Key usage: The Caesar Cipher uses the simplest possible key — a single integer shift value applied uniformly to every letter of the alphabet, giving only 25 usable keys. The Monoalphabetic Cipher generalizes this to a full, arbitrary one-to-one mapping of all 26 letters, giving a theoretical key space of $26!$ (about 4×10^{26}) possible substitution tables. The Vigenère Cipher moves to a polyalphabetic scheme: a repeating keyword determines a different Caesar-style shift for each letter position, so the effective key is the keyword itself, and its length directly affects security. The Hill Cipher moves to a fundamentally different, algebraic key: an $n \times n$ matrix of integers (modulo 26) that must be invertible modulo 26, and encryption/decryption are expressed as matrix multiplication rather than simple letter substitution.

Complexity: Complexity increases steadily across the four techniques. The Caesar Cipher requires only simple modular addition and is complex enough for a person to compute by hand almost instantly. The Monoalphabetic Cipher requires memorizing or referencing an arbitrary 26-symbol mapping table, which is more effort but still simple modular substitution. The Vigenère Cipher requires tracking a repeating keyword alongside the plaintext and applying a different shift per position, introducing a time/position-dependent element absent from the first two. The Hill Cipher requires genuine linear algebra — matrix multiplication for encryption and modular matrix inversion for decryption — making it by far the most mathematically demanding of the four to perform by hand or to break.

Security strength: Despite its enormous $26!$ key space, the Monoalphabetic Cipher is only marginally stronger in practice than the Caesar Cipher, because substitution alone never disturbs the underlying letter-frequency distribution of the plaintext language — both are broken quickly using classical frequency analysis (matching ciphertext letter frequencies to known English letter frequencies, e.g., that 'E' is the most common letter). The Hill Cipher is the strongest of the four: because it operates on blocks of letters simultaneously through matrix transformation rather than one letter at a time, it destroys single-letter frequency patterns almost entirely and resists classical frequency analysis very effectively; its main weakness is instead a known-plaintext attack, where an attacker who obtains matching plaintext/ciphertext letter-blocks can solve a system of linear equations to recover the key matrix directly.

(d) Which Technique Is Most Secure, and Why

Among the four techniques compared, the Hill Cipher is the most secure. Its security advantage comes from a structural difference rather than merely a larger key space: Caesar, Monoalphabetic, and (to a lesser extent) Vigenère are all essentially letter-by-letter substitutions, so statistical properties of the source language leak through into the ciphertext and can be exploited by frequency-based cryptanalysis. The Hill Cipher instead encrypts blocks of multiple letters together as vectors, transformed by matrix multiplication, which mixes the statistical contribution of several plaintext letters into each ciphertext block. This block-wise mixing is conceptually the classical-cryptography ancestor of the 'diffusion' property that modern block ciphers like AES rely on, and it is precisely what makes the Hill Cipher resistant to simple frequency analysis in a way that none of the other three techniques achieve.

It is important to note, however, that 'most secure among these four classical techniques' is a relative, historical comparison — the Hill Cipher is still a linear transformation and is completely broken by a known-plaintext attack using basic linear algebra, and none of these four classical substitution ciphers would be considered secure by modern standards. All of them are vulnerable to computational brute-force or systematic mathematical attack and have long been superseded in real-world use by modern symmetric algorithms such as AES and asymmetric algorithms such as RSA, which combine both substitution (confusion) and transposition (diffusion) across many rounds with vastly larger key spaces

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies.	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation